

Shoaib Zafar

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Address: Via Bruno Schreiber, 9, 43124, Parma, Italy (Home) |

Address: Via Giuseppe Moruzzi, 1, 56124 Pisa PI, 56124, Pisa, Italy (Work)

ABOUT ME

I have done Ph.D. in Emerging Digital Technologies from Scuola Superiore Sant'Anna, Pisa, specializing in functional safety and secure communication for industrial Cyber-Physical Systems. My doctoral research centered on the openSAFETY protocol, where I developed the first analytical framework for timing-parameter configuration, improving operational availability and safety assurance in safety-critical networks. Before my Ph.D., I worked in industrial automation, intelligent transportation systems, and university teaching, experience that keeps my research grounded in practical engineering constraints rather than purely theoretical problems. I am now seeking postdoctoral, research scientist, or lecturer opportunities in academia or industry, with interests spanning Industrial IoT, AI applied to IoT, real-time systems, and industrial communication safety and security.

EDUCATION AND TRAINING

01/10/2022 – 30/09/2025 Pisa, Italy

PHD IN EMERGING DIGITAL TECHNOLOGIES (EMBEDDED SYSTEMS) Scuola Superiore Sant'Anna Pisa

During my Ph.D., I investigated functional-safety communication in industrial Cyber-Physical Systems, with a particular focus on the openSAFETY protocol over heterogeneous networks, including wired Ethernet and IEEE 802.11. I conducted a detailed study of openSAFETY communication behavior under realistic network conditions and introduced an analytical approach validated through real-time experiments. A central part of my research addressed time synchronization and its related effects, which are essential for safety-related communication in the openSAFETY/openBUS standard. I developed a measurement-driven algorithm to systematically configure time-synchronization and validation parameters, enabling more predictable and repeatable commissioning under network impairments, and proposed parameter-derivation methods that improved determinism, reliability, and operational availability compared with ad hoc configuration. I also designed a lightweight AI- and trust-based resilience mechanism to detect stealthy delay-based and resource-degrading Denial-of-Service attacks in safety-critical communication, strengthening system robustness while preserving the protocol's black-channel safety assumptions.

Website <https://www.santannapisa.it/it> |

Field of study Inter-disciplinary programmes and qualifications involving Information and Communication Technologies (ICTs), Software and applications development and analysis

Level in EQF EQF level 8 |

Thesis Towards Safe, Reliable, and Secure Communication for Industrial Cyber-Physical Systems using openSAFETY Protocol

31/08/2019 – 30/07/2021 Harbin, China

MASTERS OF ENGINEERING IN COMPUTER SCIENCE AND TECHNOLOGY Harbin Institute of Technology China (Italian equivalence of foreign title is available)

Masters research in real-time embedded systems, focusing on service quality, timing analysis, and power optimization in battery-powered and safety-critical applications. The work proposed MODE and RaSAM for reducing power consumption in multiprocessor systems using dynamic power management and voltage scaling with Global EDF scheduling, and introduced a method for measuring worst-case response time in periodically triggered automotive task-chains using compositional performance analysis (pyCPA) in Python.

Address Harbin Institute of Technology 11, Siling Str, Harbin, China, Harbin Institute of Technology 11, Siling Str, Harbin, China, 150001, Harbin, China

Website www.hit.edu.cn | **Field of study** Real-time Embedded Systems, Computer vision and Deep-learning | **Final grade** 3.85/4 |

Thesis Energy-Efficient Computation and Modeling Advancements in Multi-Processor Scheduling for Real-time Embedded Systems

02/02/2010 – 02/02/2014 Lahore, Pakistan

BACHELORS OF SCIENCE IN ELECTRICAL (COMPUTER) ENGINEERING COMSATS institute of Information technology, Lahore

Analysis of dynamic voltage and frequency scaling strategies in real-time embedded systems on an operating system level to reduce the power consumption in real-time embedded systems, implemented on STORM (Simulation tool on real-time multiprocessor systems).

Address 1.5 KM Defence Rd, off Raiwand Road, Lda Avenue Phase 1 Lda Avenue, Lahore, 1.5 KM Defence Rd, off Raiwand Road, Lda Avenue Phase 1 Lda Avenue, Lahore, 54000, Lahore, Pakistan

Website <https://lahore.comsats.edu.pk/> | **Field of study** Real-time Embedded Systems | **Final grade** 3.26/4 |

Thesis Analysis of Dynamic voltage and frequency scaling for parallel homogeneous/heterogeneous embedded real-time multiprocessor architecture

13/02/2014 – 17/04/2014 Lahore, Pakistan

CCNA (TRAINING) Corvit Lahore

Address 11 A, D-1, Ghalib Rd, Block D 1 Gulberg III, Lahore, Punjab, 11 A, D-1, Ghalib Rd, Block D 1 Gulberg III, Lahore, Punjab, 54000, Lahore, Pakistan

Website <http://www.corvit.com/systems/>

13/03/2014 – 05/06/2014 Lahore, Pakistan

RHCE (TRAINING) Corvit Lahore

Address 11 A, D-1, Ghalib Rd, Block D 1 Gulberg III, Lahore, Punjab, 11 A, D-1, Ghalib Rd, Block D 1 Gulberg III, Lahore, Punjab, 54000, Lahore, Pakistan

Website <http://www.corvit.com/systems/>

23/03/2014 – 27/04/2014 Lahore, Pakistan

CCNA SECURITY (TRAINING) Corvit Lahore

Address 11 A, D-1, Ghalib Rd, Block D 1 Gulberg III, Lahore, Punjab, 11 A, D-1, Ghalib Rd, Block D 1 Gulberg III, Lahore, Punjab, 54000, Lahore, Pakistan

Website <http://www.corvit.com/systems/>

PUBLICATIONS

2025

[Time synchronization and performance analysis of the openSAFETY protocol via UDP over Ethernet](#)

<https://doi.org/10.1016/j.sysarc.2025.103605>

Authors: Shoaib Zafar, Salvatore Sabina, Alessandro Biondi, Giorgio Buttazzo | **Journal Name:** Journal of Systems Architecture | **Volume, Issue and Pages:** Volume 169, 2025, 103605, | **Publisher:** Elsevier

2026

[Preliminary experimental evaluation of time synchronization parameters in openSAFETY over IEEE 802.11 wireless networks](#)

DOI: [10.1109/WFCS67029.2026.11511646](https://doi.org/10.1109/WFCS67029.2026.11511646)

Authors: Shoaib Zafar, Alessandro Biondi, Salvatore Sabina, Giorgio Buttazzo | **Journal Name:** 2026 IEEE 22nd International Conference on Factory Communication Systems (WFCS) | **Publisher:** IEEE

Link <https://ieeexplore.ieee.org/document/11511646>

2026

[A Trust-Based DoS Resilience Framework for the openSAFETY Domain](#)

The Manuscript is Under-Review

Authors: Shoaib Zafar, Dimitrios Pliatsios, Panagiotis Sarigiannidis, Salvatore Sabina, Giorgio Buttazzo

2022

[A Survey on MAC-Based Physical Layer Security over Wireless Sensor Network](#)

<https://doi.org/10.3390/electronics11162529>

Authors: Attique Ur Rehman, Muhammad Sajid Mahmood, Shoaib Zafar, Muhammad Ahsan Raza, Fahad Qaswar, Sumayh S. Aljameel, Irfan Ullah Khan, Nida Aslam | **Journal Name:** Electronics | **Volume, Issue and Pages:** Electronics, 11(16), 2529 | **Publisher:** MDPI

<http://dx.doi.org/10.4108/eetcasa.v8i1.2546>**Authors:** M.H. Zafar, A.U. Rehman, S. Zafar | **Journal Name:** EAI Endorsed Transactions on Context-Aware Systems and Applications | **Publisher:** EAI

WORK EXPERIENCE

VISITING RESEARCHER – UNIVERSITY OF WESTERN MACEDONIA – 01/04/2025 – 30/09/2025 – KOZANI, GREECE

Business or Sector: Information and communication |**Department:** Internet of Things & Applications Lab (ITHACA), Department of Electrical and Computer Engineering |**Website:** <https://ithaca.ece.uowm.gr/>

- Designed and implemented a trust-based DoS resilience framework for the openSAFETY protocol, targeting stealthy delay-based and resource-degrading attacks in safety-critical industrial networks.
- Conducted real-time experimental evaluation of the framework on a physical openSAFETY testbed, demonstrating reliable attack detection while preserving black-channel safety assumptions.
- Co-authored a paper (under review, 2026).

ENGINEERING LECTURER – LAHORE GARRISON UNIVERSITY – 18/10/2021 – 04/10/2022 – LAHORE, PAKISTAN

Business or Sector: Education | **Department:** Department of Software Engineering |**Address:** Lahore Garrison University DHA Phase 6, Sector C. Lahore, Lahore Garrison University DHA Phase 6, Sector C. Lahore, Lahore, Pakistan |**Website:** www.lgu.edu.pk

- Designed and delivered undergraduate courses in Real-Time Embedded Systems, Algorithms, Data Structures, and Operating Systems
- Supervised final-year undergraduate dissertations and led a research group of undergraduate research assistants
- Authored research publications and prepared grant applications to secure external fundingManaged departmental responsibilities including student admissions, induction programs, and committee participation

INTELLIGENT TRANSPORTATION SYSTEM ENGINEER – CHINA STATE CONSTRUCTION AND ENGINEERING CORPORATION LIMITED – 24/10/2017 – 20/08/2019 – MULTAN, PAKISTAN

Business or Sector: Construction | **Department:** Intelligent Transportation Systems |**Address:** China State Construction Engineering Corporation Limited. Plot 360, Industrial Estate Multan, China State Construction Engineering Corporation Limited. Plot 360, Industrial Estate Multan, 60000, Multan, Pakistan**Website:** <https://csci.cscec.com/en/xwzx/xw/201911/2992228.html>

- Deployed and integrated intelligent transportation systems on the Multan–Sukkur motorway, including automatic incident detection, license plate recognition, and AI-based speed monitoring
- Verified real-time performance of safety-critical ITS applications under operational highway conditions
- Awarded Outstanding Pakistani Employee (2018) with a \$1,000 cash prize for project performance

PROFESSIONAL PROJECT:

CSCEC-PKM PROJECT (Multan-Sukkur) Section.

ELECTRICAL ENGINEER – VOLKA FOOD INTERNATIONAL – 13/05/2015 – 17/10/2017 – MULTAN, PAKISTAN

Business or Sector: Manufacturing

- Designed, installed, and commissioned electrical and automation systems for industrial food processing plants using Siemens PLC and SIMATIC HMI platforms
- Developed embedded cards for real-time wrapper control in high-speed packaging machines
- Implemented PID-based automation and control solutions, improving equipment reliability and reducing operational interruptions
- Served as technical lead for installation, commissioning, and testing across multiple plant automation projects

TRAINEE ENGINEER – LAFARGE PAKISTAN – 01/03/2014 – 01/04/2015 – CHAKWAL, PAKISTAN

Business or Sector: Construction

- Supported electrical preventive and corrective maintenance activities for industrial plant system.

PROJECTS

02/02/2024 – 30/09/2025

Systematic Timing Configuration and Resilient Safety Communication for Industrial Cyber-Physical Systems using openSAFETY Protocol

During the PhD, functional-safety communication in Industrial Cyber-Physical Systems was investigated, with a special focus on the openSAFETY protocol operating over heterogeneous networks, including wired Ethernet and IEEE 802.11. This work represents the first comprehensive in-depth study on this topic, and, for the first time, an analytical approach was presented and validated through real-time experiments. Especially, the time synchronization and its interrelated effects analyzed in this work constitute the fundamental building blocks of safety-related communication in the openBUS standard. A measurement-driven algorithm was developed to systematically configure time synchronization and validation parameters, enabling predictable and repeatable commissioning under realistic network impairments. The approach was validated through real-time experiments, and parameter-derivation algorithms were proposed to improve determinism, reliability, and operational availability compared to ad hoc approach. In addition, a lightweight AI and trust-based resilience mechanism was designed to detect stealthy delay-based and resource-degrading Denial-of-Service attacks in safety-critical communications, thereby strengthening robustness while preserving black-channel safety assumptions.

Link <https://www.sciencedirect.com/science/article/pii/S1383762125002772>

01/02/2023 – 28/03/2023

Experimental analysis of Maximum Latency by varying the Linux kernel preemption models

As part of the Advanced Operating Systems course during my PhD, this project focused on the experimental analysis of Linux kernel latency across different kernel preemption models using the cyclictst tool. The main goal was to determine how non-preemptive, voluntary preemption, and fully preemptive real-time kernels behave under controlled CPU and I/O stress conditions, with focus on system responsiveness, determinism, and suitability for real-time applications. The project included configuring multiple kernel environments, designing stress workloads, and setting up a structured testing framework to study kernel behavior in latency-sensitive scenarios.

13/10/2023 – 18/02/2024

Bypassing the stack protection in Linux using ROP chains

As part of the System-Level Cyber Security course during my PhD, this project focused on low-level software and binary security, with particular focus on analyzing Linux memory protection mechanisms and studying how Return-Oriented Programming (ROP) can be used to bypass non-executable stack protections via carefully crafted gadget chains and system-call manipulation. The broader goal of the project was to build a practical understanding of how modern exploitation techniques operate at the system level and how such knowledge can inform stronger defensive design. In parallel, the course provided in-depth exposure to major attack and defense mechanisms, including buffer overflows, heap overflows, return-to-libc, ROP, speculative execution attacks, Rowhammer, and state-of-the-art microarchitectural attacks, establishing a strong foundation in both offensive security analysis and system-level protection strategies.

06/08/2023 – 13/11/2023

Digital Twin of Double Pendulum in constrained cyber-physical systems

As part of the Cyber-Physical Systems course during my PhD, this project focused on the design and implementation of a real-time digital twin of a double pendulum using the Ptask library in C. The main goal was to model the dynamics of a nonlinear physical system within a real-time computing framework, combining physical modeling, task-based system design, and interactive simulation. The project involved developing user-controlled simulation parameters, defining shared data structures, and implementing dedicated real-time tasks for state updating and graphical rendering, with the objective of studying how timing, control flow, and system behavior can be integrated in a cyber-physical application

31/08/2019 – 30/06/2021

Worst-case estimation on latency bound for task-chains in real-time embedded systems

Completed as my Master's dissertation at Harbin Institute of Technology, this project investigated the problem of end-to-end timing analysis in distributed embedded systems, with particular relevance to advanced industrial control domains such as robotics and automotive applications. The study focused on task-chain-based execution models, where tasks are linked through data dependencies across multiple resources, creating interdependent challenges in response-time and communication-delay analysis. The objective of the work was to establish a holistic latency analysis approach for automotive embedded systems by examining timing behavior across the full execution path under different task-priority configurations.

31/08/2019 – 30/06/2021

Energy-Efficient computation for Real-time Multiprocessor Systems

As part of my Master's dissertation, this project investigated power-aware optimization in real-time embedded multiprocessor systems, where rising transistor density and growing application demands have made energy consumption, heat dissipation, and battery limitations critical concerns. The work focused on operating-system-level techniques to enhance system efficiency while preserving responsiveness and quality of service in constrained embedded environments. In this context, two algorithms, MODE and RaSAM, were proposed by integrating Dynamic Power Management (DPM) and Dynamic Voltage and Frequency Scaling (DVFS) within a Global EDF scheduling framework to support energy-efficient execution in multiprocessor real-time systems.

24/10/2017 – 16/08/2019

CSCEC-PKM PROJECT

Served as an Intelligent Transportation Systems Engineer on the CSCEC-PKM Project, where I contributed to the development of a central command and control center for intelligent traffic operations. My responsibilities included the deployment, integration, and real-time performance verification of incident detection systems, weather detection systems, FM radio communication systems, and

advanced intelligent safety solutions for accident prevention. This work supported the implementation of smart transportation infrastructure designed to enhance operational efficiency, traffic safety, and real-time decision-making.

Link <https://csci.cscec.com/en/xwzx/xw/201911/2992228.html>

12/05/2015 – 16/10/2017

Automation and Advancements of Food Plants and Packaging Machines

PROFESSIONAL PROJECTS

Installation, Testing, and Commissioning of MOGUL Jelly Plant

Completed the full installation and commissioning of a MOGUL jelly plant, including electrical panels and system interfacing with Siemens PLCs, servo drives, servo motors, and induction motors. Contributed to process automation and PID implementation for the starch drying panel and MOGUL production line.

- **PLC:** Siemens ET200S
- **Drives:** Siemens
- **Servo Drive:** Bosch Rexroth

Installation, Testing, Commissioning, and Automation of Jelly Kitchen

Executed the complete installation and automation of a jelly kitchen using Siemens S7-300 PLC and SIMATIC HMI. Implemented backpressure and flow control using PID valves, pressure transmitters, and RTD sensors, and designed an autonomous control system for automatic batch processing.

- **PLC:** Siemens S7-300
- **Drives:** Danfoss

Installation, Testing, Commissioning, and Automation of Center-Filled Chew Candy Plant

Led the installation, commissioning, and automation of a center-filled chew candy plant, supporting the integration of industrial control systems for reliable plant operation.

- **PLC:** Omron
- **Drives:** SEW

Installation, Testing, Commissioning, and Automation of Bubble Gum Plant

Managed the complete installation and automation of a bubble gum plant, including an extended PROFORM cooling tunnel and U1 packing machines. Designed and implemented an embedded card for real-time wrapper control of the U1 bubble packing machine.

- **PLC:** Siemens S7-400

Embedded System Design and Implementation for Real-Time Wrapper Control of Biscuit and Bubble Packing Machines

Designed and implemented embedded control systems for real-time wrapper control in biscuit and bubble packing machines, focusing on synchronization, reliability, and high-speed machine performance.

Installation, Testing, Commissioning, and Automation of Curing Rooms for HVAC Systems

Executed installation, testing, commissioning, and automation of curing rooms for heating, ventilation, and air conditioning systems, ensuring operational reliability and process control.

Installation, Testing, Commissioning, and Automation of External Pump Systems for Lithium Bromide Water Absorption Chillers

Delivered installation, automation, and commissioning support for external pump systems used in lithium bromide water absorption chiller applications, ensuring proper control integration and system functionality.

WROKSHOPS

15/12/2024 – 17/12/2024

Challenge-based human inspired computer vision- Challenge-1&2

CONFERENCES AND SEMINARS

20/04/2026 – 25/04/2026 Offenburg- Germany

IEEE International Conference of Factory Communication Systems WFCS 2026

Link <https://www.hs-offenburg.de/wfcs2026/en/program>

HONOURS AND AWARDS

01/10/2022

PhD Scholarship (PNRR) by Italian government – PNRR

31/08/2019

Full-time Chinese Government Scholarship 2019-2021 – China Scholarship Council

Full-time Chinese Government Scholarship for Masters in Engineering in Computer Science and Technology.

29/09/2018

Outstanding Pakistani Employee – China State Construction Engineering Corporation

Awarded as Outstanding Pakistani Employee for 2018 performance with \$1000 cash prize- CSCEC-PKM Project

● **SKILLS**

Industrial Protocols & Standards

openSAFETY | Communication protocols, Profinet, Profibus, Ethernet, EtherCAT, Modbus | communication protocols (Profinet, OPCUA)

Real-Time & Embedded Systems

Worst Case Analysis | linux kernel | STORM simulator for Real-Time Embedded Systems | Embedded computing | Real-time Scheduling | PyCPA | Energy Aware Embedded systems

Cybersecurity

Trust-based anomaly detection | DoS resilience | Stack based Buffer Overflow attacks | Heap overflow attacks | ROP chain exploitation | Microarchitectural Attacks and Defenses | RAS Extension for ARM processors

AI & Machine Learning

Long Short Term Memory (LSTM) | ANN/CNN/RNN & Transformers | Variational Autoencoders (VAE) | Time-Series Prediction using Deep Learning | Transfer Learning for Digital Twin Models | SHapley Additive exPlanations | machine learning | TinyML

Programming & Tools

Python | C++ | Programming C | raspberry pi | Assembly (computer programming)

● **LANGUAGE SKILLS**

Mother tongue(s): **URDU**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C1	C1	C1	C1	C1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user